

Grade 7
Unit 9
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

Determine if the expressions are equivalent for the given values.

	Value	$8x + 4x - 5$	$2(6x - 8) + 11$	Equivalent?
1.	$x = -4$	$8(-4) + 4(-4) - 5$ $-32 + (-16) - 5$ $-48 - 5$ -53	$2(6(-4) - 8) + 11$ $2(-24 - 8) + 11$ $-48 + (-16) + 11$ $-64 + 11$ -53	☒ yes ☐ no
2.	$x = 0$	$8(0) + 4(0) - 5$ $0 + 0 - 5$ -5	$2(6 \cdot 0 - 8) + 11$ $2(0 - 8) + 11$ $2(-8) + 11$ $-16 + 11$ -5	☒ yes ☐ no
3.	$x = 0.5$	$8(0.5) + 4(0.5) - 5$ $4 + 2 - 5$ 1	$2(6 \cdot 0.5 - 8) + 11$ $2(3 - 8) + 11$ $2(-5) + 11$ $-10 + 11$ 1	☒ yes ☐ no
4.	$x = \frac{1}{3}$	$8\left(\frac{1}{3}\right) + 4\left(\frac{1}{3}\right) - 5$ $\frac{8}{3} + \frac{4}{3} - 5$ $\frac{12}{3} - 5$ $4 - 5$ -1	$2\left(6 \cdot \frac{1}{3} - 8\right) + 11$ $2\left(\frac{6}{3} - 8\right) + 11$ $2(2 - 8) + 11$ $2(-6) + 11$ $-12 + 11$ -1	☐ yes ☒ no

5. Depending on the value being substituted into the expressions, the evaluated

expressions

☒ always
☐ sometimes
☐ never

had the same value as each other.

Determine whether the expressions are equivalent for the given values.

		$\frac{1}{5}(15 + 20y) + 2x$	$4\left(2x + y + \frac{3}{4}\right)$	Equivalent?
6.	$x = -2$ $y = 6$	$\frac{1}{5}(15 + 20 \cdot 6) + 2(-2)$ $\frac{1}{5}(15 + 120) - 4$ $\frac{1}{5}(135) - 4$ $27 - 4 = 23$	$4\left(2(-2) + 6 + \frac{3}{4}\right)$ $4\left(-4 + 6 + \frac{3}{4}\right)$ $4(2.75)$ 11	☐ yes ☒ no
7.	$x = 0$ $y = 1$	$\frac{1}{5}(15 + 20 \cdot 1) + 2 \cdot 0$ $\frac{1}{5}(15 + 20) + 0$ $\frac{1}{5}(35) + 0$ 7	$4\left(2 \cdot 0 + 1 + \frac{3}{4}\right)$ $4\left(0 + 1 + \frac{3}{4}\right)$ $4(1.75)$ 7	☐ yes ☒ no
8.	Answers will vary $x = 2$ $y = 0$	$\frac{1}{5}(15 + 20 \cdot 0) + 2(2)$ $\frac{1}{5}(15 + 0) + 4$ $\frac{1}{5}(15) + 4$ $3 + 4 = 7$	$4\left(2 \cdot 2 + 0 + \frac{3}{4}\right)$ $4\left(4 + 0 + \frac{3}{4}\right)$ $4(4.75)$ 19	☐ yes ☒ no

Determine if the following algebraic expressions are equivalent using any method you choose.

	Expression A	Expression B	Equivalent?
9.	$x(13 - 15) - \frac{3}{2}(4 + 6x)$ $-2x \quad -6 - 9x$ $-11x - 6$ $-11(0) - 6$ -6	$-11x - 15$ $-11(0) - 15$ -15	☐ yes ☒ no
10.	$\frac{5}{7}(2x + 12x) - (6^2)$ $\frac{5}{7}(14x) - 36$ $10x - 36$ $10(0) - 36$ -36	$-36 + 10x$ $-36 + 10(0)$ $-36 + 0$ -36	☐ yes ☒ no